Portfolio

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01

Reflow

Mar–May 2025

A hydrogen fuel-cell bus system that captures water already produced onboard and reintegrates it into cooling, cleaning, and daily operations—reducing waste through measurable, service-ready design.

*Project developed under the RISD × HMG Collaborative Regeneration Program.

Opportunity / Water Recovery

Fuel-cell operation creates water. Fleet operations can put that byproduct to work.

9 L

of water per 1 kg of H_2 consumed

Normally discarded; recoverable for bus washing and cooling.

DAILY H_2 / BUS	WATER / BUS	100-BUS FLEET / DAY
10 kg	90 L	9,000 L
20 kg	180 L	18,000 L
30 kg	270 L	27,000 L

Water-stressed operations / Los Angeles: ~10% annual water loss through leaks and evaporation. Dubai: desalination at high energy cost.

Operational perspectives

The system must be measurable, maintainable, scalable and understandable.

TRANSIT OPS MANAGER

Sofia Ramirez

Director of Bus Operations – Arid Metro

Reduce costs and meet sustainability targets; scale fleetwide without schedule disruption or new-system downtime.

MAINTENANCE TECHNICIAN

Derek Hall

Senior Tech – Central Depot

Keep buses reliable; fit inspection and draining into daily checks; avoid difficult plumbing, leaks, freeze risks and longer service.

PASSENGER / NGO ADVOCATE

Sofia Wong

Community Sustainability Liaison

Make environmental impact visible and transparent; build public trust with evidence rather than hidden “green” claims.

Water already exists around and within the vehicle → capture and reuse it.

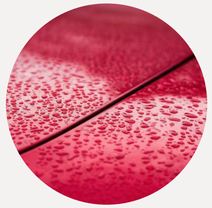
Latent Water Resources

Unused Byproducts in Bus Systems, Especially Hydrogen Fuel Cell Buses

Collect Onboard Water

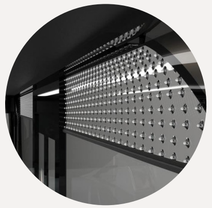
Capture and Repurpose Efficiently

Inner window fog

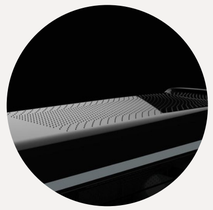


Dewdrops / condensation

Fuel-cell water emission



Inner window



Roof

Water tank

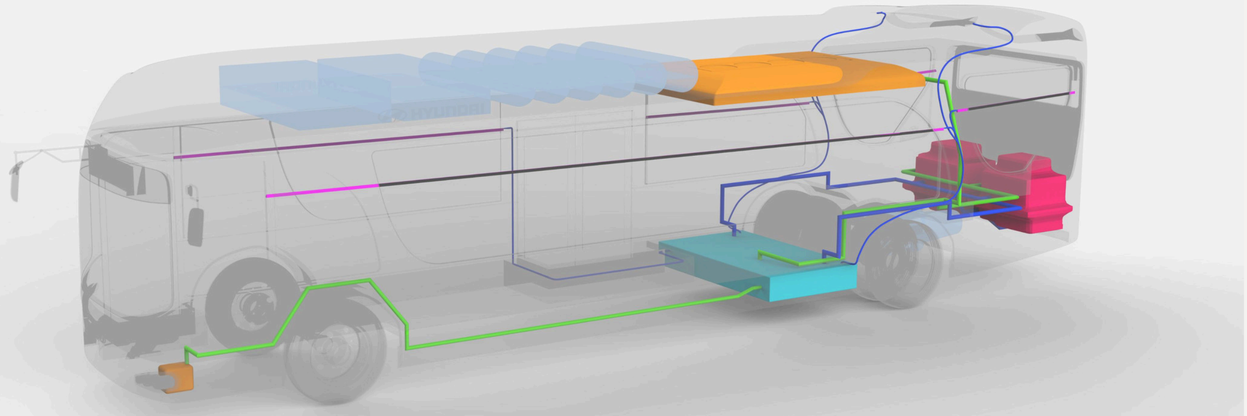
System Concept / Flow

SYSTEM FLOW

- **Capture surfaces** (roof bumps, interior window channels, exhaust condenser)
- **Debris mesh** (200–500 µm)
- **UV-C chamber** (LED 265–280 nm)
- **Distribution manifold** (solenoid valves)
- **Reuse endpoints** (washer jets, condenser spray, cleaning hose)
- **Storage tank** (80–120 L)
- **Cyclone trap**
- **Gutters & downcomers** (gravity-fed)

Physical bus integration

Operational Flow of Onboard Water Capture and Reuse



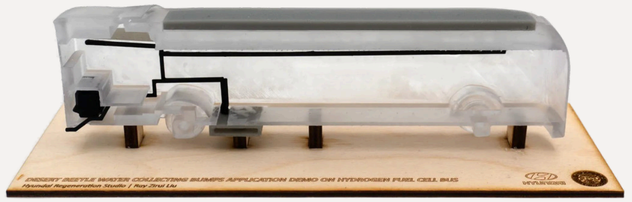
- Inside cabin window water collection directing channels
- Water collection inputs (roof, inside cabin, water emission)
- Water tank
- Water repurpose outputs
- Receival systems (thermal, AC, battery, windshield fluid)
- Fuel cell engine systems

Operational Flow

1. Condense: Droplets form on bump surface / exhaust condenser
2. Collect: Micro-gutters route water to downcomers
3. Filter & Sanitize: Mesh → cyclone → UV-C
4. Store: Tank level tracked; overflow routed to safe drain
5. Reuse: Driver selectable or automatic: washer / cooling mist / depot hose
6. Log: Volume in/out recorded; faults raised to dash + service app

Internal Water Collecting Flow Demonstration

1:53 physical demonstration model



Biomimicry / Surface Development

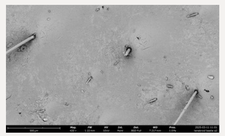
Biological inspiration → observation → geometry translation → fabrication → controlled testing

01 Biological inspiration

Certain darkling beetles in the Namib Desert use their bodies as fog collectors through fog-basking behavior.

02 Microscope + SEM

Microscope

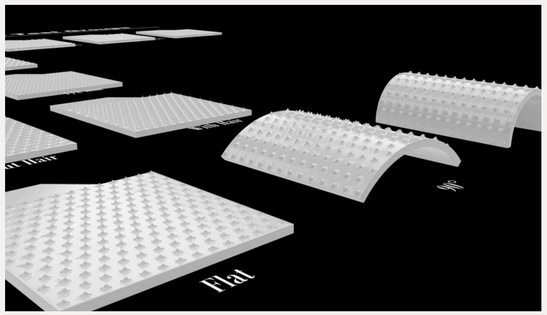


SEM

5 darkling beetle samples studied under microscope and SEM to analyze biological surface structures.

Research reference: Hunter King's team at the University of Akron found that a sphere with 1 mm bumps captured droplets at nearly 2.5× the efficiency of a smooth sphere with the same surface area.

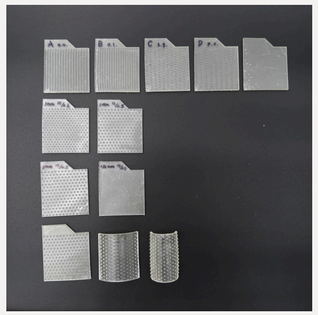
03 Geometry translation



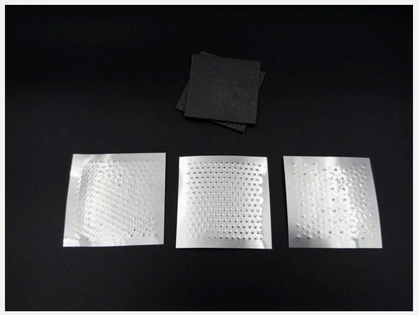
3D-modeled bump samples

Types · Scales · Density · Curvature · Hair Effect

04 Fabrication



3D printed samples

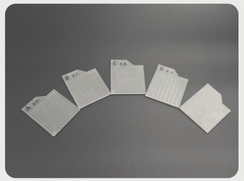


Stamped metal samples

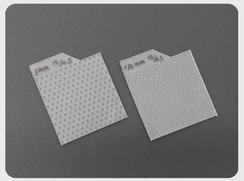
20+ modular bump samples developed and fabricated.

05 Controlled variable experiment

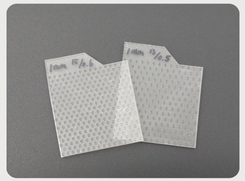
`Boiling water → 45° downward → 1¼ in above water level → 2 min → compare W₀ and W₁`



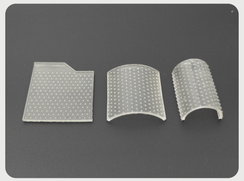
Types



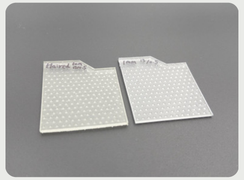
Scales



Density



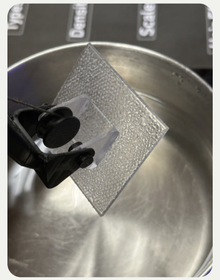
Curvature



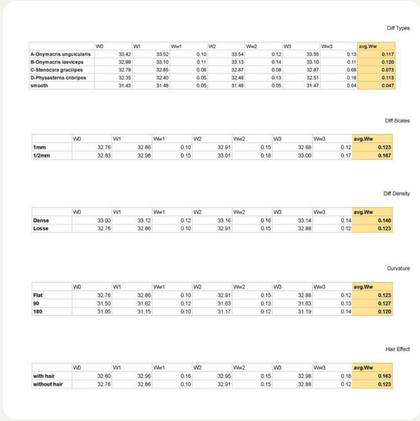
Hair Effect

5+ test groups / 10+ test samples

Controlled comparisons of surface configurations to evaluate water-capture efficiency.



Controlled capture test



Diff Types

	W0	W1	Ww1	W2	Ww2	W3	Ww3	avg.Ww
A-Onymacris unguicularis	33.42	33.52	0.10	33.54	0.12	33.55	0.13	0.117
B-Onymacris laeviceps	32.99	33.10	0.11	33.13	0.14	33.10	0.11	0.139
C-Stenocara gracilipes	32.79	32.85	0.06	32.87	0.08	32.87	0.08	0.073
D-Physosterna cribripes	32.35	32.40	0.05	32.48	0.13	32.51	0.16	0.113
smooth	31.43	31.48	0.05	31.48	0.05	31.47	0.04	0.047

Diff Scales

	W0	W1	Ww1	W2	Ww2	W3	Ww3	avg.Ww
1mm	32.76	32.86	0.10	32.91	0.15	32.88	0.12	0.123
1/2mm	32.83	32.98	0.15	33.01	0.18	33.00	0.17	0.167

Diff Density

	W0	W1	Ww1	W2	Ww2	W3	Ww3	avg.Ww
Dense	33.00	33.12	0.12	33.16	0.16	33.14	0.14	0.140
Loose	32.76	32.86	0.10	32.91	0.15	32.88	0.12	0.123

Curvature

	W0	W1	Ww1	W2	Ww2	W3	Ww3	avg.Ww
Flat	32.76	32.86	0.10	32.91	0.15	32.88	0.12	0.123
90	31.50	31.63	0.12	31.63	0.13	31.63	0.13	0.127
180	31.05	31.15	0.10	31.17	0.12	31.19	0.14	0.120

Hair Effect

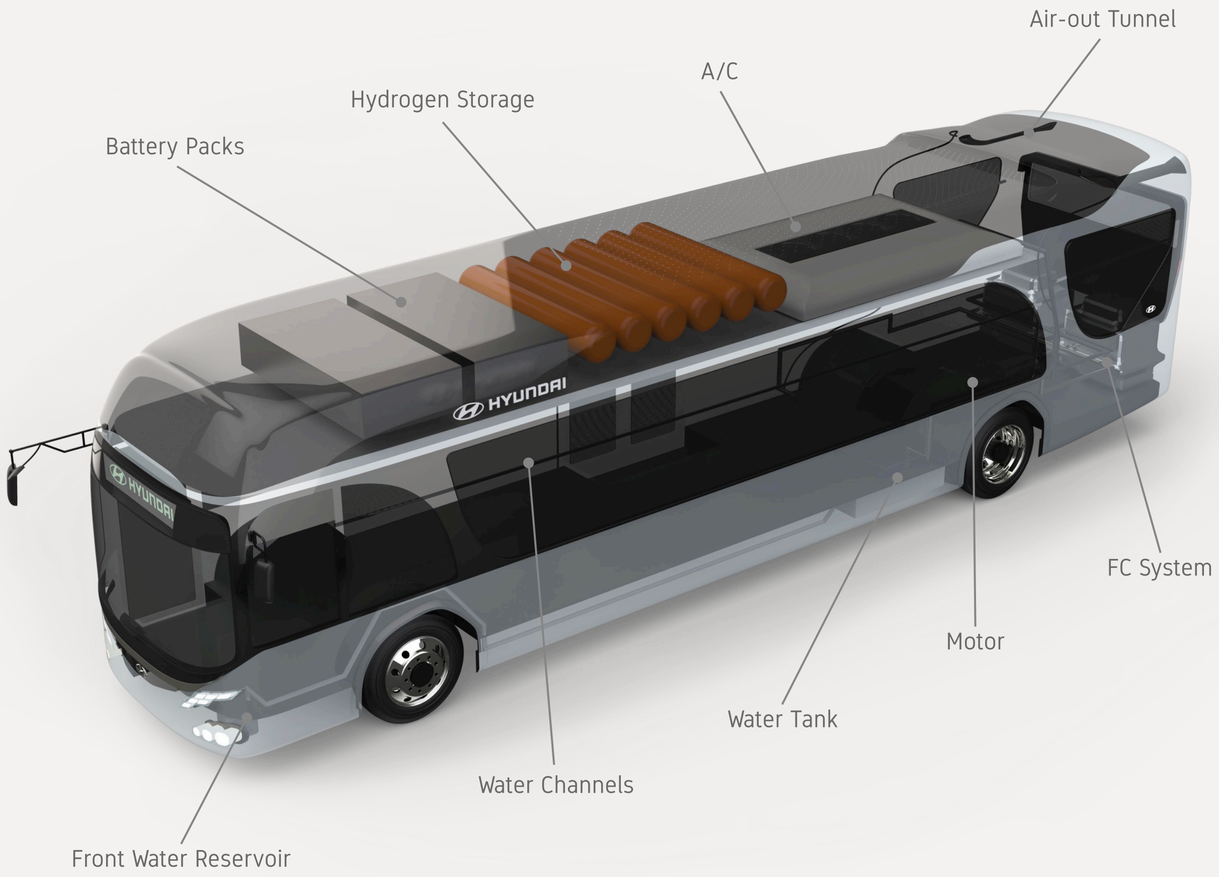
	W0	W1	Ww1	W2	Ww2	W3	Ww3	avg.Ww
with hair	32.80	32.96	0.16	32.95	0.15	32.96	0.16	0.163
without hair	32.76	32.86	0.10	32.91	0.15	32.88	0.12	0.123

W_0 / W_1 measurement record

Application / Vehicle Integration

Tested surface geometry becomes part of the bus architecture and onboard water system.

INTEGRATED VEHICLE

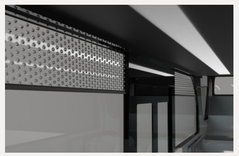


Redesigned exterior and frame carry treated surfaces, collection hardware, routed water and storage.

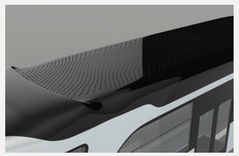
Application map

01 APPLIED SURFACES

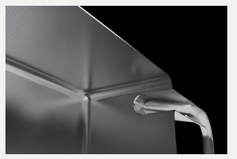
Inner Window Surfaces



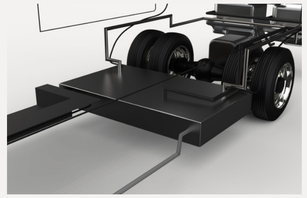
Roof Surface and Roof-end Water Collector



Inner Surface of Water Tank



02 COLLECTION + STORAGE



Water Tank

03 PRACTICAL REUSE

- Water assisting cooling system (evaporative or coils/pipes) for AC, fuel cell and battery
- Water-spray cooling system on radiator fans
- Windshield washing system
- Depot cleaning

*Bump geometry has been intentionally scaled up to illustrate function more clearly within modeling constraints.
Ideal bump size: < 0.5 mm

Operations / HMI / Service Logic

Reflow turns live system data and service records into decisions for operators, fleet staff and technicians.

OPERATIONAL SERVICE LOOP

01	→ 02	→ 03	→ 04	→ 05	→ 06
Monitor live status	**Review history**	**Detect abnormality**	**Diagnose issue**	**Service or adjust**	**Return to operation**
Tank, water volume, mode	Use, recovery, fault record	Warning or deviation	Locate fault and service need	Inspect, repair, reset	Confirm normal state
OBSERVE			INTERPRET		ACT + VERIFY

CONDITION → ACTION

OK	Continue operation
Warning / due soon	Inspect or monitor soon
Fault / overdue	Trigger maintenance action

Examples: filter warning · UV off · low tank sensor error

THREE VIEWS OF THE SAME OPERATING SYSTEM

DRIVER / OPERATOR

Home Dashboard

Tank level, reclaimed/used water, mode and status.

FLEET / OPERATIONS STAFF

Logs Page

Review daily performance, fault events, notes and CSV history.

← Bus #27 — Export CSV

Date	Reclaimed (L)	Used (L)	Fault Events	Notes
Aug 21	78	65	None	Normal operation
Aug 20	72	54	Filter warning	Filter cleaned 08/21
Aug 19	83	60	UV off (10 min)	Auto reset successful
Aug 18	75	62	None	Normal operation
Aug 17	70	59	Low tank sensor error	Cleared after restart
Aug 16	80	66	None	Normal operation

7-day reclaimed average: 76 L — 7-day used average: 61 L — Fault count (past week): 3

MAINTENANCE TECHNICIAN

Maintenance Page

Prioritize tasks by interval, status, due date and service history.

← Bus #27 – Maintenance Status

Task	Interval	Status	Due Date / Countdown	Notes
Replace Mesh Filter	14 days	Due soon	In 6 days (Aug 28)	Last replaced Aug 14
Run UV Test	30 days	OK	In 27 days (Sep 18)	Auto-test passed Aug 18
Flush Tank	90 days	OK	In 74 days (Nov 4)	No issues detected
Inspect Roof Panels	180 days	OK	In 142 days (Jan 12)	No cracks / seal secure
Replace Pump Cartridge	365 days	OK	In 320 days (Jul 14)	Installed new unit Aug 1
Update Firmware (Control)	180 days	OK	In 150 days (Jan 20)	Last update Jul 22

Mark Completed — View History — Parts Needed

Green (OK) → >14 days away (safe).
Yellow (Due soon) → ≤7 days away.
Red (Overdue) → missed deadline.

Integrated Prospect

A vision of the bus operating within everyday city transit.

02

Genesis

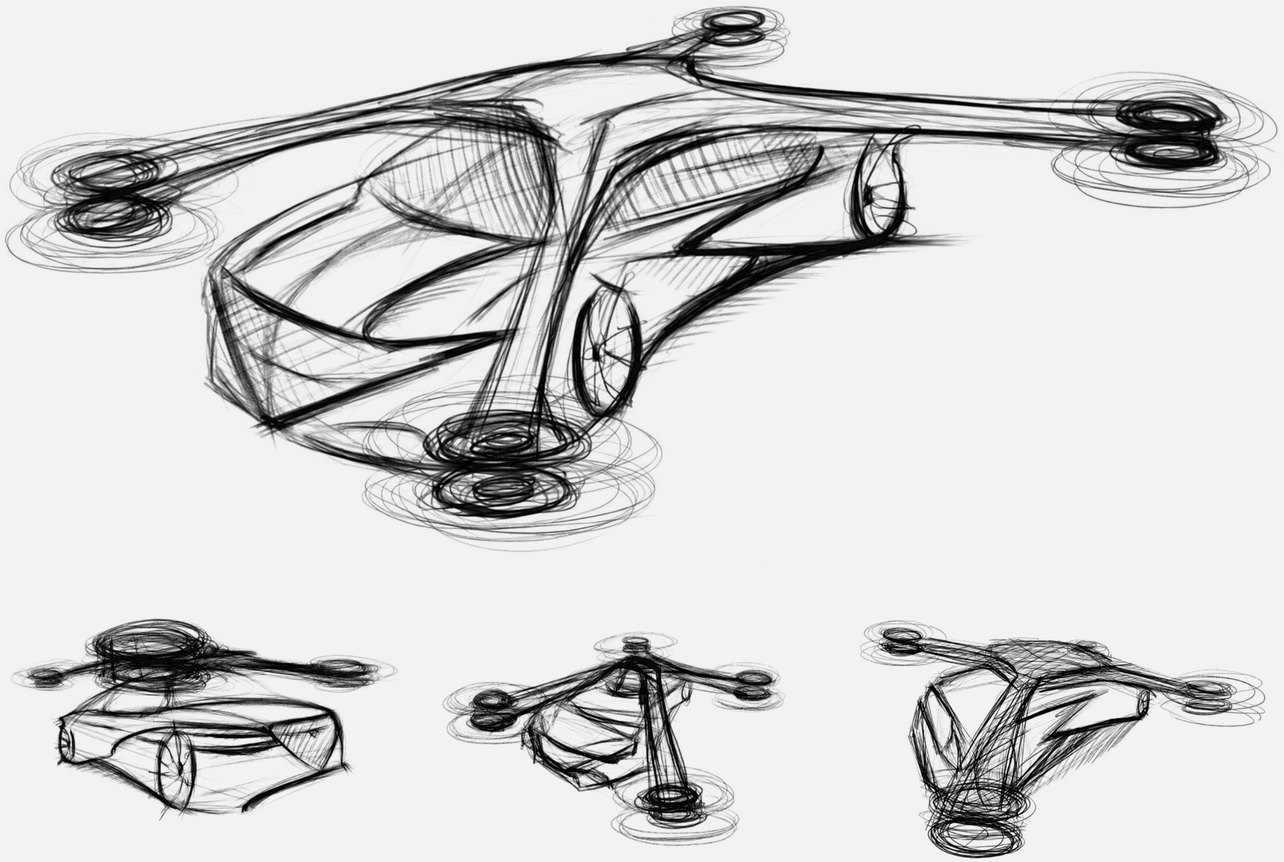
Nov–Dec 2024

A flying mobility concept exploring how Genesis automotive design language can translate into an aerial vehicle through deployable propulsion, aerodynamic surfacing, and continuous exterior form.

Form / Architecture Development

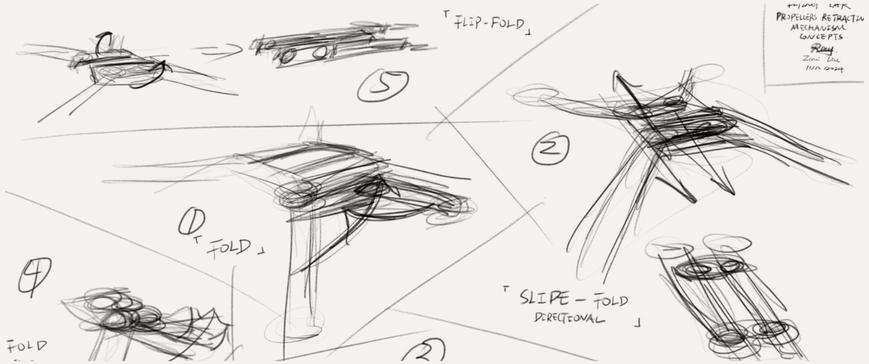
VEHICLE ARCHITECTURE

Proportion and packaging studies explored how an automotive body could accommodate an aerial propulsion system.



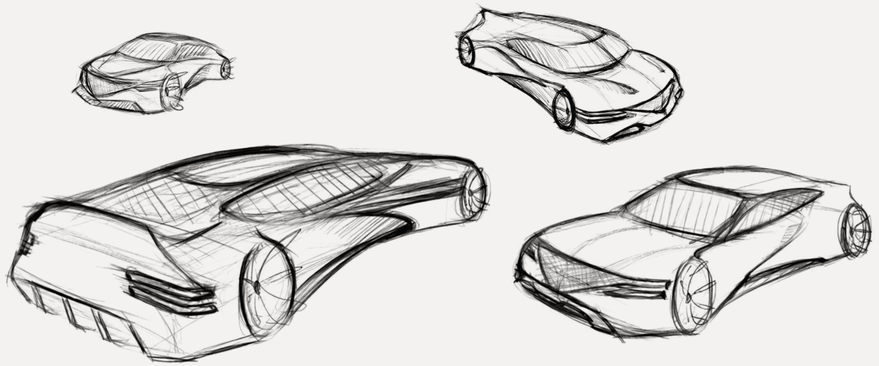
AERIAL MECHANISM

Rotor placement and folding concepts developed the transition between road and flight configurations.

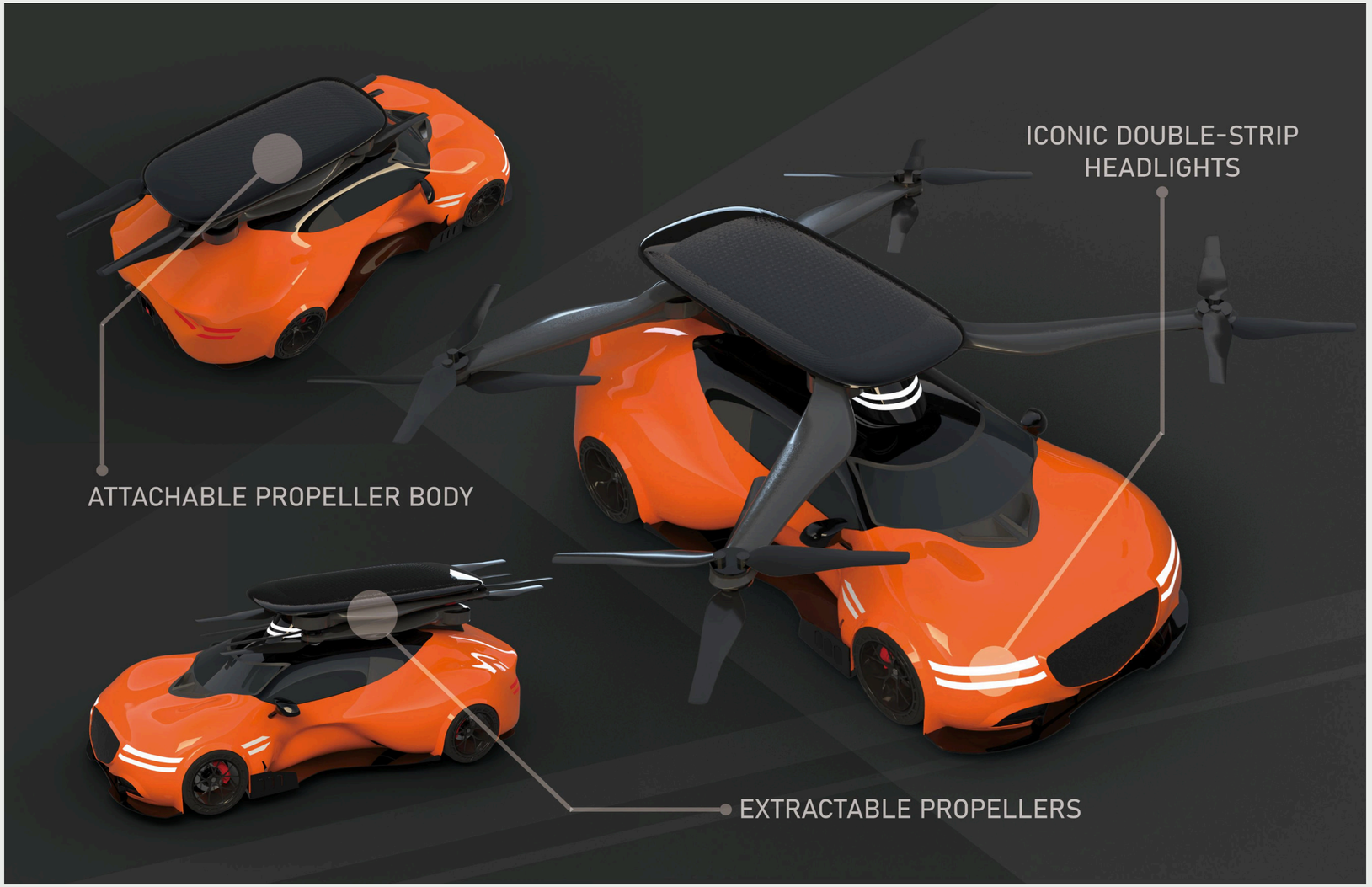


GENESIS DESIGN LANGUAGE

Lighting, proportion and surface cues were translated into the new vehicle architecture.



Aerial System Integration



Dual-mode architecture

ROAD CONFIGURATION ↔ AERIAL CONFIGURATION

Automotive body + deployable aerial propulsion.

CONTEXT

An exploration within urban aerial mobility and the growing eVTOL design space.

DESIGN GAP

Aircraft-led concepts leave room for a more recognizable luxury automotive identity.

DESIGN INTENT

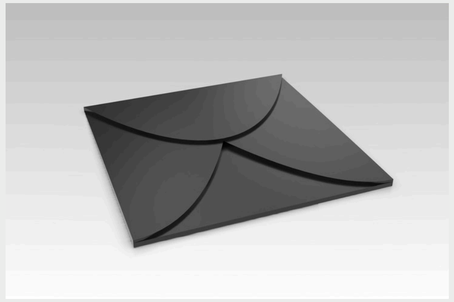
Translate Genesis identity into an aerial mobility architecture.

Biological Reference / Surface Translation

BIOLOGICAL REFERENCE → SURFACE ABSTRACTION → VEHICLE APPLICATION

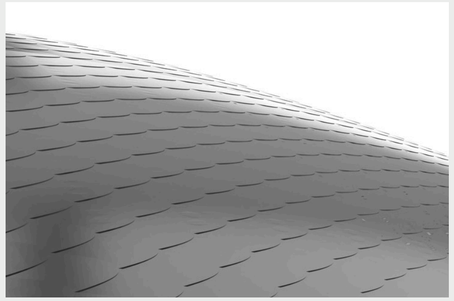
Layered overlap

Overlapping fish scales informed a layered, repeatable exterior surface treatment.



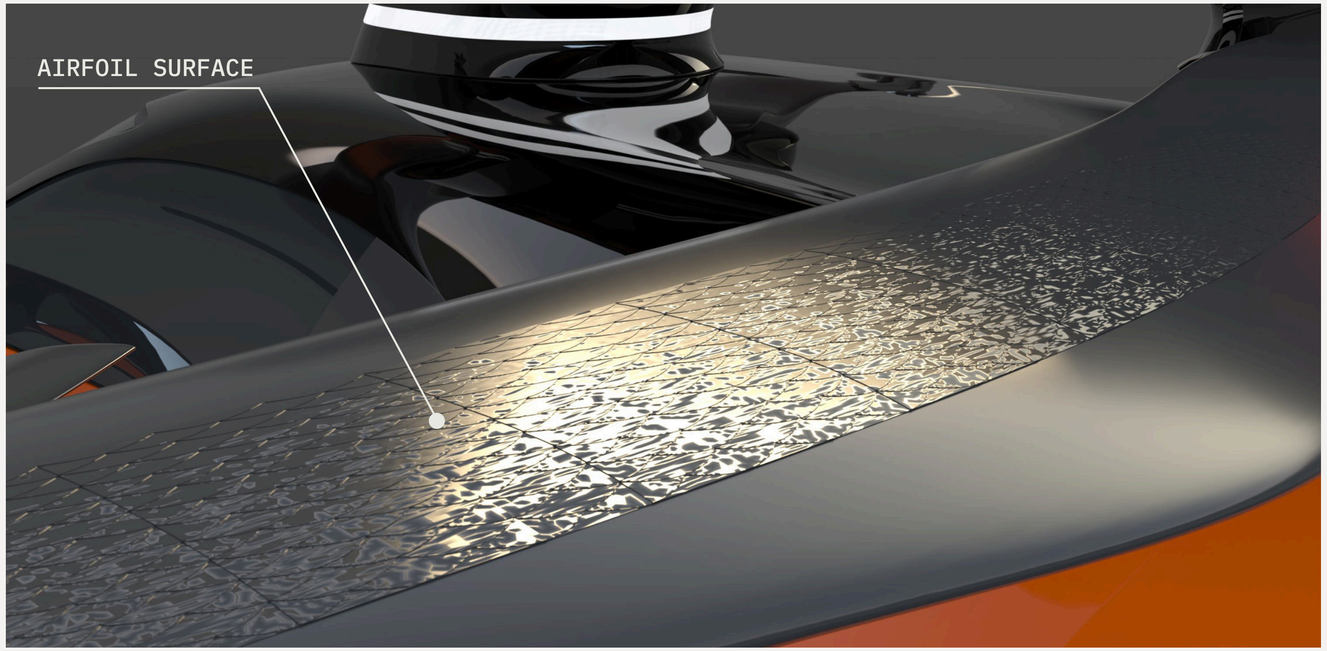
SCALE MODULE

A scalloped edge becomes a repeatable surface unit.



CURVED SURFACE STUDY

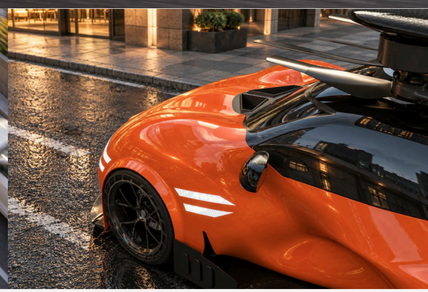
Repeated relief follows a continuous curved surface.



Roof / Propeller Body

The scale-derived relief is applied selectively to the roof / propeller body and airfoil surfaces.

Urban Mobility Vision



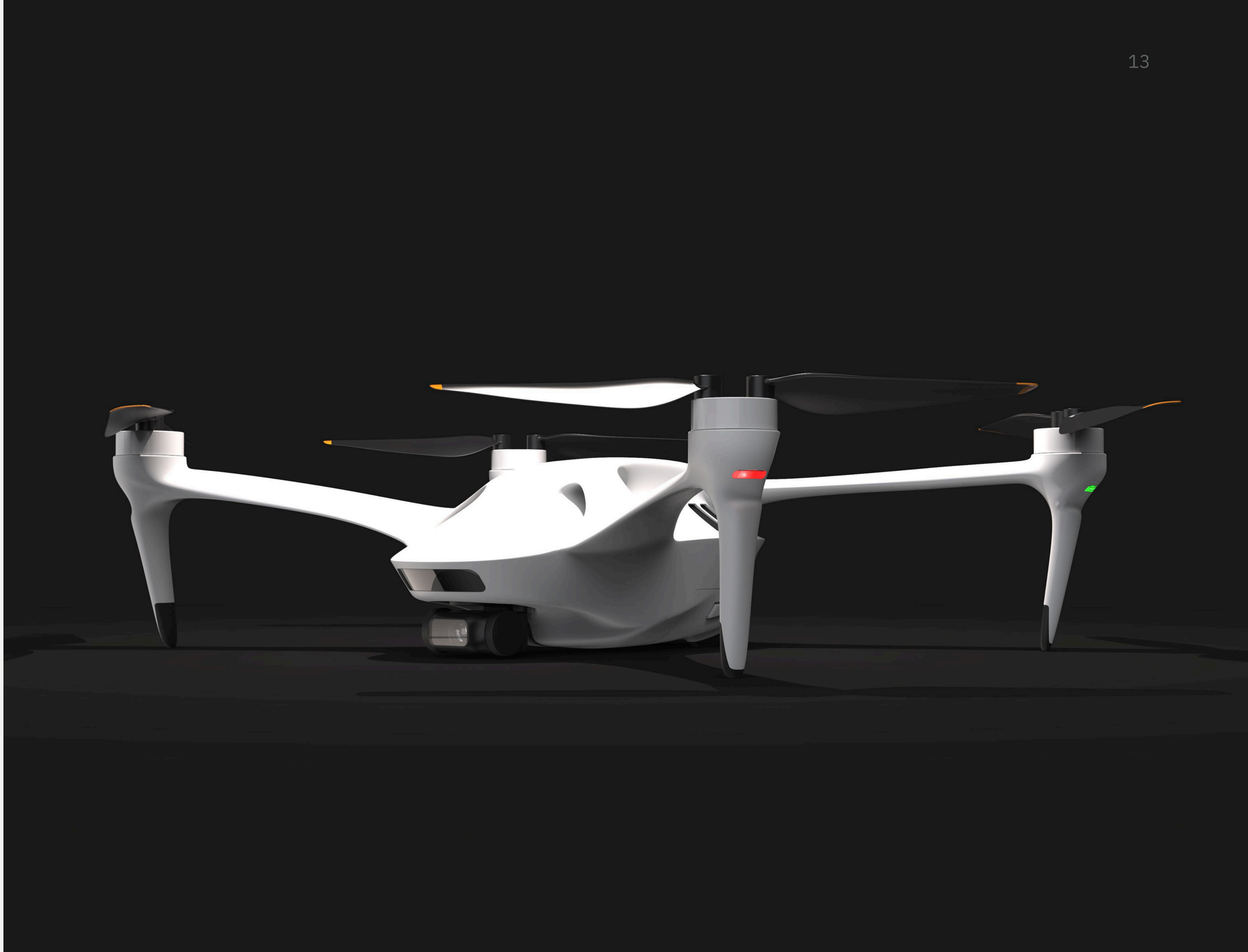
A dual-mode mobility concept translating Genesis automotive identity into both road and aerial contexts.

03

Echo

Feb–Mar 2024; Oct–Nov 2025

A drone form study exploring aerodynamic structure, portability, and visual identity through reference-driven form development.

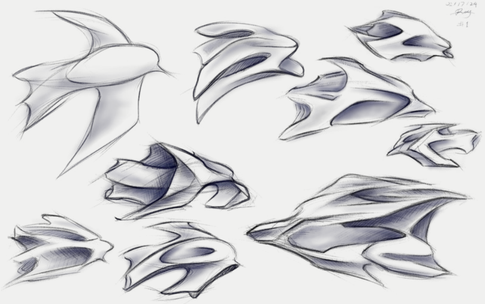


Reference / Form Translation

Three real-world references were abstracted into distinct form cues, then translated through sketch exploration into the drone’s proportion, surfaces, and structural character.

SEA BIRD

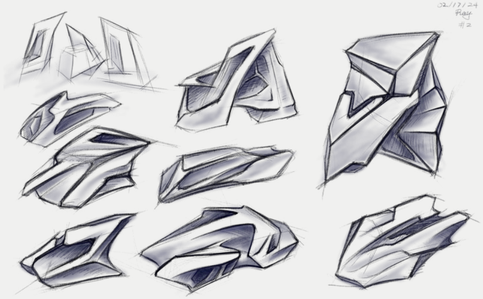
Organic flow / wing-body continuity



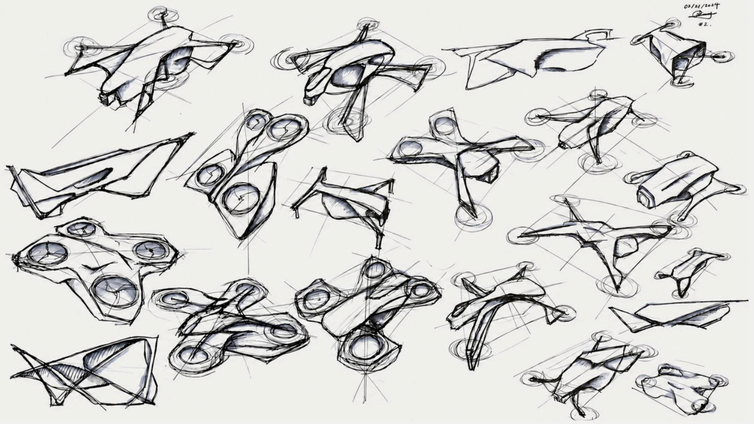
REFERENCE → SKETCH EXPLORATION

ZAHA HADID TEA AND COFFEE SERVICE SET

Sculptural geometry / sharp surface transitions

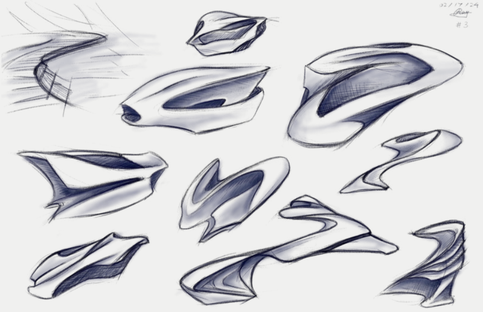


REFERENCE → SKETCH EXPLORATION

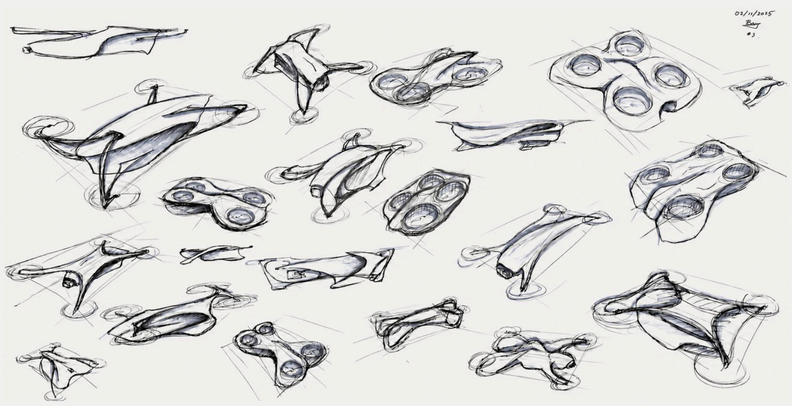


MICHAEL S. VAN LEESTEN MEMORIAL BRIDGE

Structural continuity / spanning form



REFERENCE → SKETCH EXPLORATION



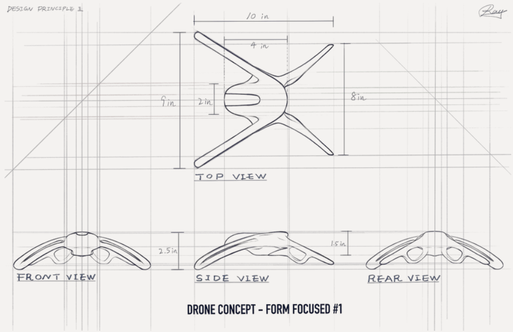
Physical Form Exploration

Sketch language → quick physical studies → resolved directions → technical form definition

SEA BIRD

Organic / agile / biomimetic

TECHNICAL FORM DEFINITION

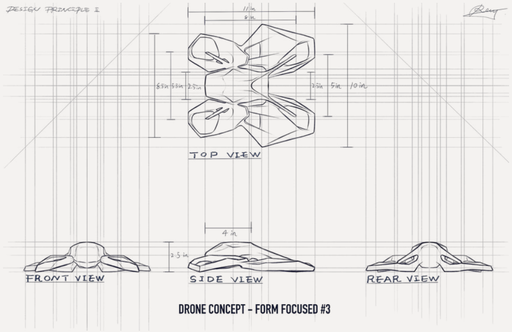


DRONE CONCEPT - FORM FOCUSED #1

ZAHA HADID TEA AND COFFEE SERVICE SET

Sharp / rigid / geometric

TECHNICAL FORM DEFINITION



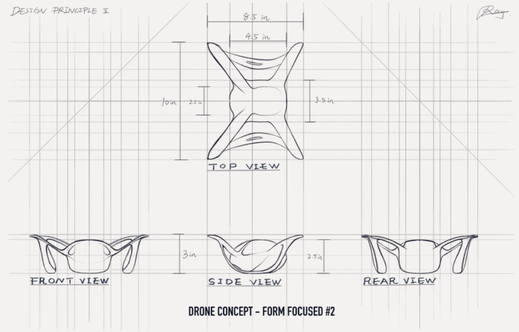
DRONE CONCEPT - FORM FOCUSED #3

MICHAEL S. VAN LEESTEN MEMORIAL BRIDGE

Curvilinear / structural / sculptural

*Selected for further development

TECHNICAL FORM DEFINITION



DRONE CONCEPT - FORM FOCUSED #2

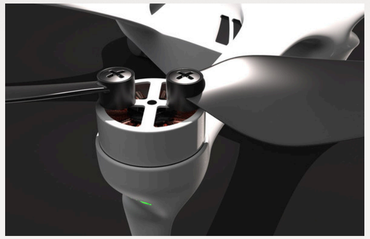
Form Resolution

STANCE / BODY–ARM CONTINUITY

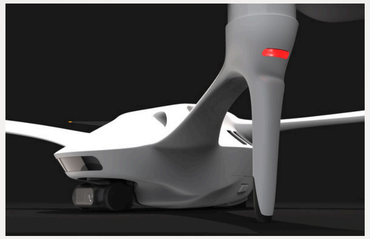
The selected direction offers stronger volumetric continuity, a clearer body-to-arm relationship, calmer surfaces, and a more legible silhouette.

DETAIL DEVELOPMENT

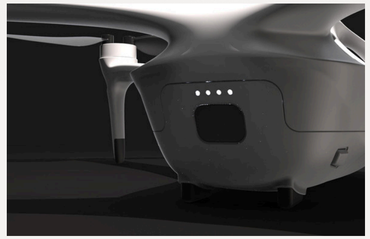
Camera / sensor housing



Rotor / arm joint



Body / opening



Rear interface

Flight Context

The resolved form is presented across varied flight conditions to show its silhouette, proportion, and visual identity in motion.

04

Traverse

Apr–May 2024; Jul–Aug 2025

A snow-terrain EMS vehicle designed to bridge the gap between manual sled rescue and helicopter evacuation, prioritizing patient stability, crew workflow, and real alpine constraints.

Research / Rescue Context

Snow rescue must protect the patient while overcoming terrain, exposure and access limits.

01 / Reach

Deep snow, steep slopes and narrow trails restrict approach.

02 / Wait

Cold exposure continues while poor visibility complicates location and access.

03 / Evacuate

Uneven terrain adds jolts and effort; weather can ground aerial support.

For the injured patient

Cold exposure while waiting and in transit.
Snow contact, jolts and vibration stress injuries.
Isolation, panic and disorientation add distress.
Basic sleds lack insulation and stabilization.

For the rescuers

Deep snow, slopes and narrow trails limit access.
Gear hauling and unstable footing hinder response.
Poor visibility and unreliable location cues slow approach.
Ground extraction is exhausting when aircraft cannot fly.

Rescue methods compared

Higher bars indicate more time, cost or operational difficulty.

Relative levels adapted from original project research; response times are not measured.
Proposed bars show design targets, not validated performance. Helicopter access is weather-dependent.

58.4%
of recorded alpine-skiing cases required emergency-department treatment

404
reported hospital admissions in the alpine-skiing dataset

74.8%
of hospital admissions were associated with fractures

Source: PHAC / CHIRPP · Alpine skiing · 5,434 cases · Oct 2016–Jan 2024 · Sentinel emergency-department sample; not population prevalence.

Users / Design Criteria

Patient safety, crew capability and fleet reliability define the brief.

User perspective	Rescue scenario	Key needs
Alex Bergstrom 38 / Ski Patrol Responder Colorado Rocky Mountains	Remote double-black-diamond ridge; suspected pelvic fracture. With helicopters grounded and the slope too steep for a snowmobile stretcher, responders manually stabilized and moved the patient.	Stable rescue on extreme terrain Secure, isolated stretcher transport Weatherproof gear access Less physical burden during extraction
Mia Chen 24 / Intermediate skier Visiting from California	An ice patch caused an injury on an ungroomed backcountry trail. The route was too narrow for large rescue vehicles and unstable for helicopter access.	Fast, safe extraction Minimal vibration and jolting Warmth and shelter while waiting / in transit Less movement with limb or spinal injuries
Rocky Ridge Ski Patrol Command Unit Sandra Löfgren Operations Director	“We’re constantly balancing response speed, staff safety, and equipment reliability. If a vehicle can cut rescue time and reduce injury risk—while staying serviceable in harsh conditions—it earns its place in our fleet.”	Modular and reliable vehicle Reduced rescue time; low maintenance Easy operation for seasonal staff Standard EMS equipment and stretchers Reliable in extreme alpine conditions

Design criteria

01 **Stable patient transport over rough snow terrain**

02 **Access to narrow, steep, or difficult rescue environments**

03 **Protection from cold, vibration, and secondary injury**

04 **Efficient operation by a small rescue crew**

05 **Compatibility with standard EMS equipment and stretcher systems**

06 **Reliable, serviceable construction for harsh alpine conditions**

07 **Ability to support helicopter handoff when needed**

Operational use case

Deep snow, steep slopes and low visibility; where unstable sleds or road ambulances fall short.

Fractures, spinal injuries and hypothermia; maintain warmth, isolation and vibration control.

Full ground transport or a mid-route handoff to helicopter when needed.

Inspiration / Ideation

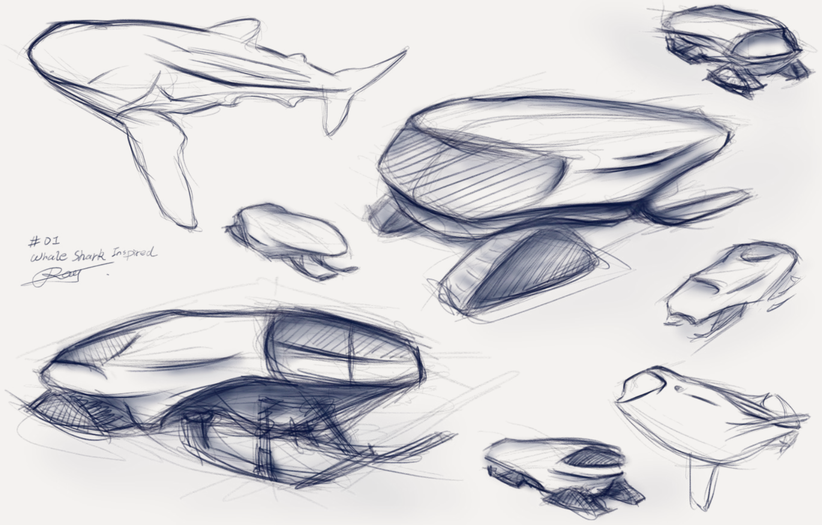
From visual references to sketch exploration and early physical studies.

01 / Inspiration

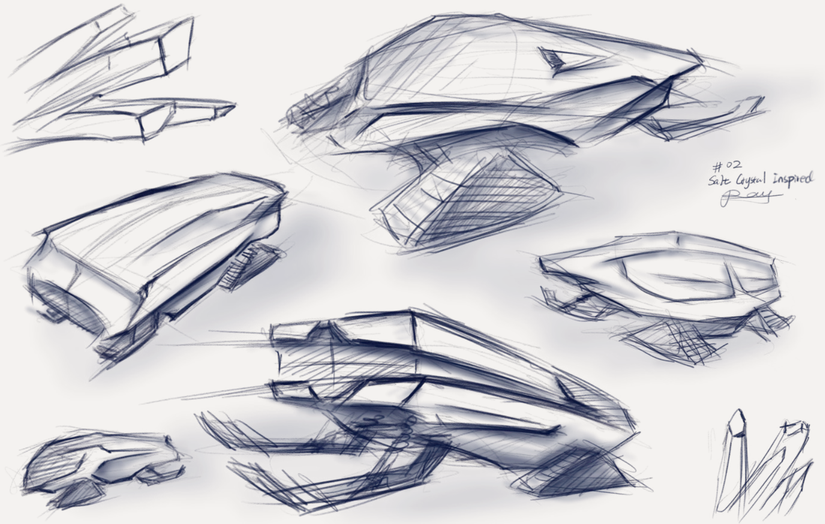
Design language

- Uniform
- Simple
- Nimble
- Dynamic

02 / Sketch exploration



Volume and stance studies

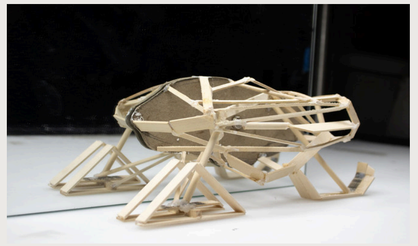


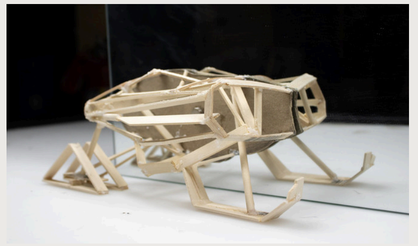
Body and silhouette studies

03 / Physical development

Layout exploration

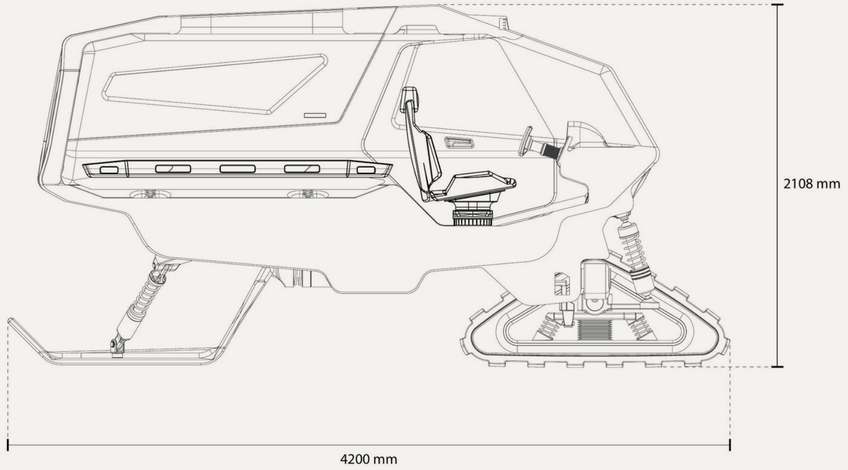
Form exploration



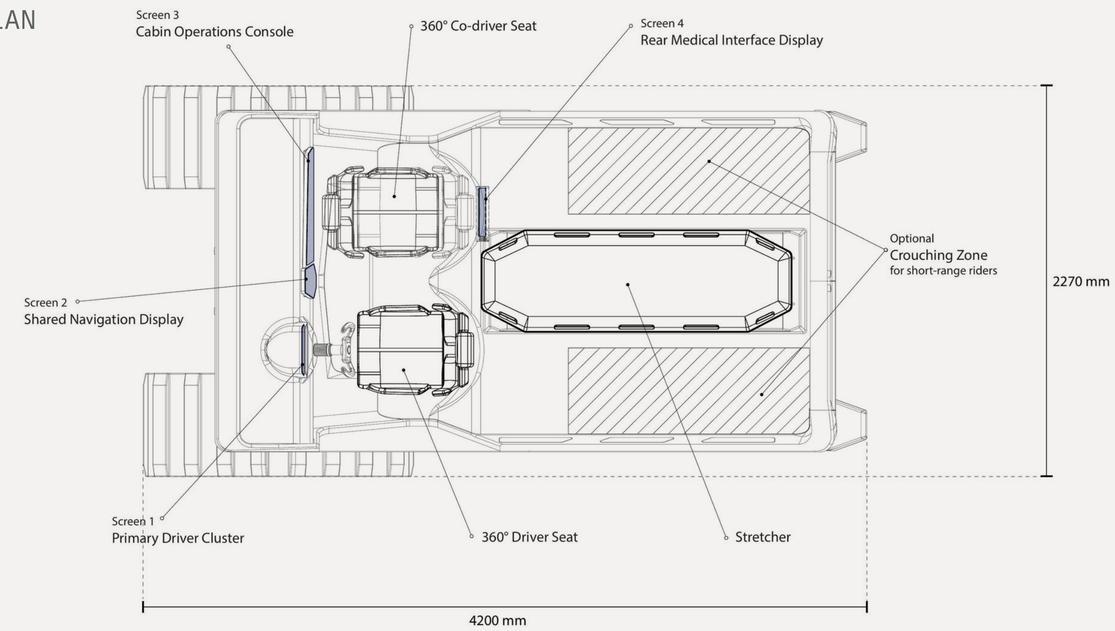


Vehicle Architecture / Packaging

01 / SIDE ELEVATION



02 / CABIN PLAN



VEHICLE ENVELOPE

4200 mm / overall length
2108 mm / overall height
2270 mm / overall width

CABIN INDEX / TOP VIEW

Screen 1	Primary Driver Cluster
Screen 2	Shared Navigation Display
Screen 3	Cabin Operations Console
Screen 4	Rear Medical Interface Display

360° Driver + Co-driver Seats

Patient Longitudinal Stretcher

Optional Crouching Zone

TWO-PERSON EMS CREW

Driver / Operator

Navigate complex terrain · Communicate with base / helicopter · Maintain route safety

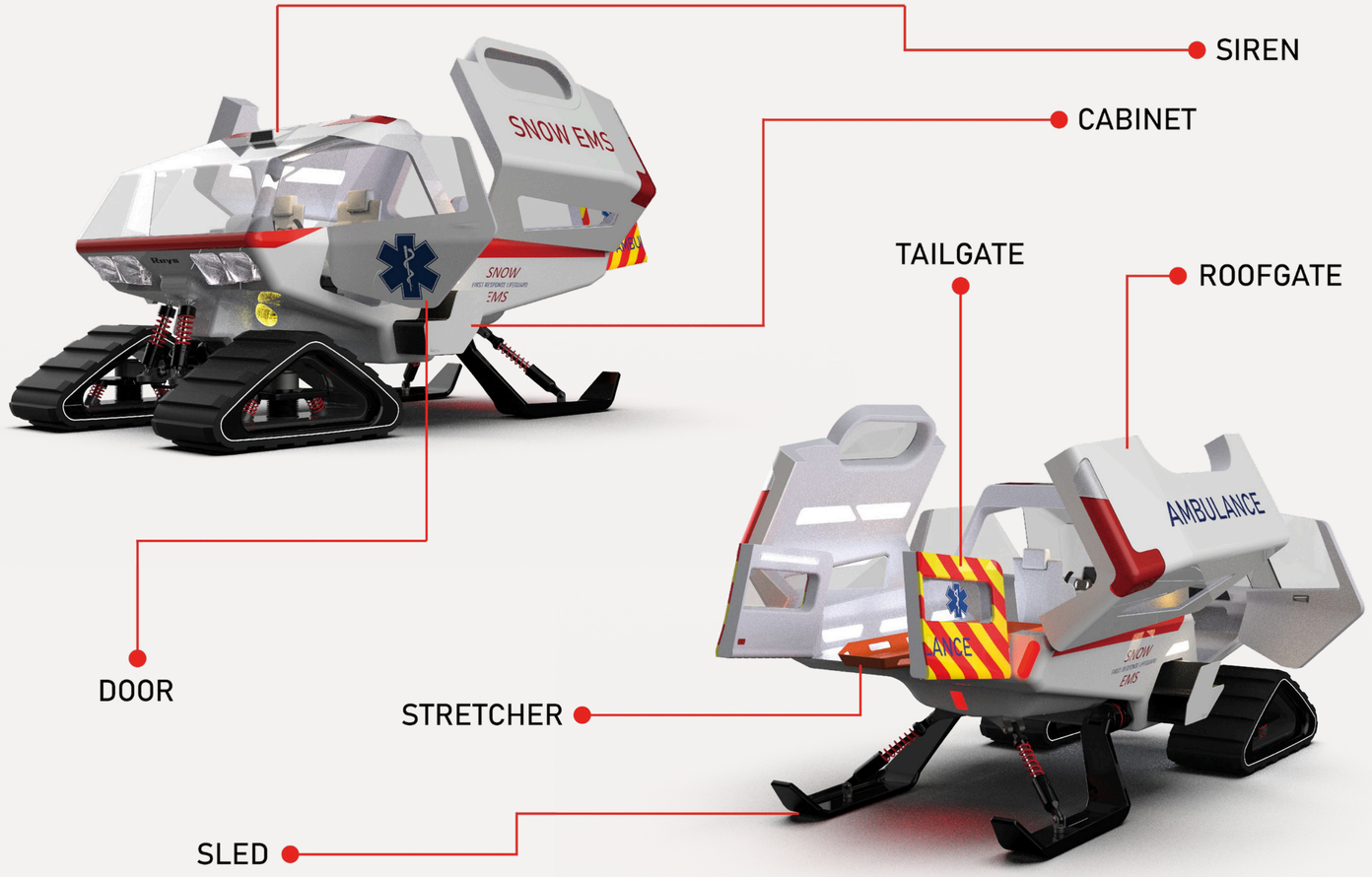
Medical Co-Driver

Monitor and stabilize the patient · Operate the medical interface · Assist with stretcher extraction or aerial handoff

Features / Rescue Operation

Key systems and access points support stable transport, crew care, and aerial transfer when conditions allow.

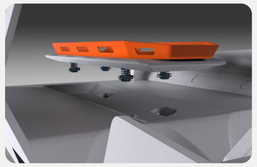
01 / CONFIGURATION



02 / HELICOPTER HANDOFF

When terrain permits, a helicopter can hover over the vehicle and lift the stretcher directly, reducing ground transport time.

03 / CORE SYSTEMS

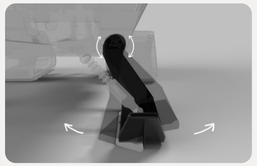


Stretcher damper

Four coilover dampers beneath the stretcher rails absorb shocks and reduce secondary injury over rough terrain.

360° crew seats

Rotatable driver and co-driver seats let responders turn toward the patient and improve access during transport.

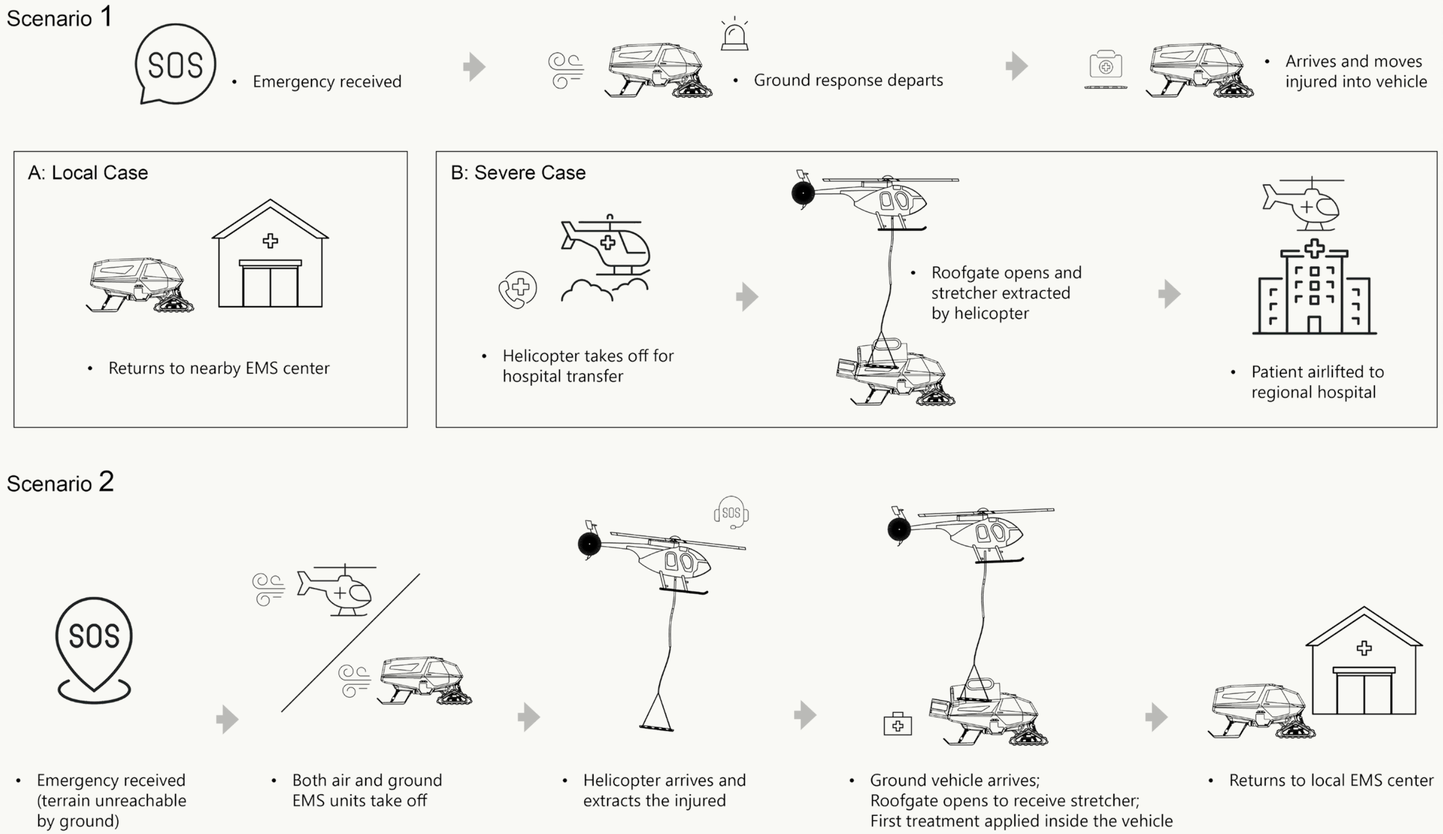


Swiveling sleds

Powered sleds adjust during steering or on extreme terrain to maintain ground contact, stability, and control.

Operational Flow / Human-centered System

01 / OPERATIONAL FLOW



02 / FUNCTIONAL + HUMAN-CENTERED DESIGN

The roof of the Snow EMS vehicle is carefully designed as both a visual tool for aerial teams and a psychological aid for patients inside.

03 / LIVERY DESIGN

Regional visibility cues

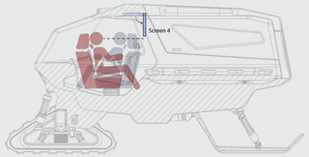
YELLOW BASE
European / Québec high-visibility practice

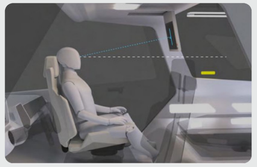
WHITE BASE
U.S. snow EMS practice in Colorado and Vermont

HMI / Interface Design

COCKPIT CONTEXT / Four interfaces across the driver and medical crew positions

ERGONOMICS & PLACEMENT

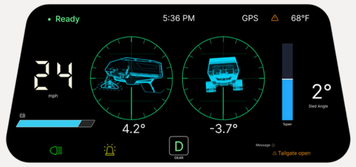




TILT / GAZE
Determined by the co-driver's relaxed gaze and perpendicular sightline. Minimizes overhead or window reflections and supports comfortable eye movement without neck strain.

HORIZON / AWARENESS
Offset above eye level without blocking the external horizon. The responder can monitor slope, daylight, helicopter arrival, and other environmental cues while reading data.

01 DASH CLUSTER



11.6 in (295 mm) W × 5.4 in (136 mm) H

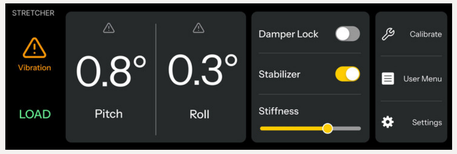
ROLE Primary driver instrument display.
PURPOSE Provides the driver with real-time performance, stability, and system status information for safe terrain navigation.

02 NAVIGATION & AIR SYSTEM

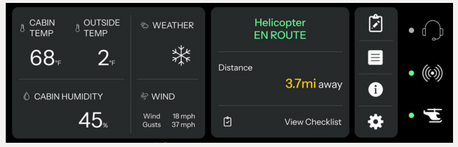
5.0 in (128 mm) W × 5.0 in (128 mm) H

ROLE Shared between driver and co-driver.
PURPOSE Combines route guidance with climate controls, enabling quick adjustments without leaving the navigation view.

03 NON-DRIVER OPS, ENVIRONMENT AND CABIN OPERATION



9.7 in (245 mm) × 3.2 in (80 mm)



8.9 in (225 mm) × 2.8 in (70 mm)

ROLE Co-driver's operations console.
PURPOSE Centralizes environmental, stability, and cabin management, allowing the co-driver to handle operations while the driver focuses on vehicle control.

04 CO-DRIVER REAR-FACING MEDICAL INTERFACE

11.0 in (280 mm) W × 7.9 in (201 mm) H

ROLE Dedicated patient monitoring and care support display for the medical co-driver.
PURPOSE Enables the co-driver to continuously monitor and manage patient condition, environmental status, and vehicle stability without distraction from driving tasks, even in rough terrain.

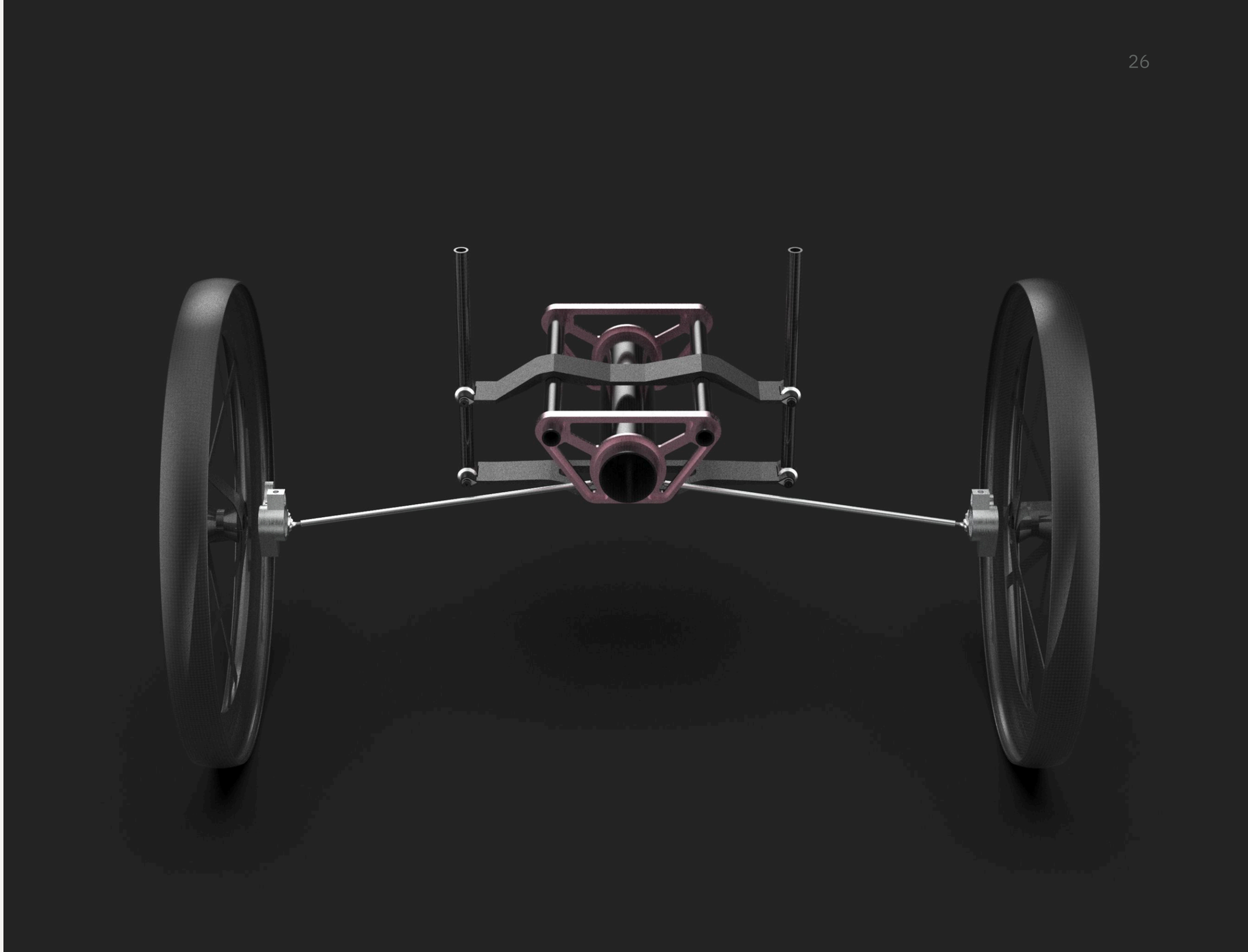
05

HERC

Rover Steering System

Sep 2025 – Apr 2026

Designing and building the complete steering system for the 2026 NASA HERC rover, from geometry and CAD/prototyping to carbon-fiber fabrication, testing, and integration. Focused on mechanical precision, durability, and fabrication feasibility.



Challenge / Role / Steering Architecture

A team-built rover for the **2026 NASA Human Exploration Rover Challenge**, developed within the RISD Rover team.
My subsystem focus: the front-end steering system.

MY ROLE & SCOPE

Steering System Technical Lead

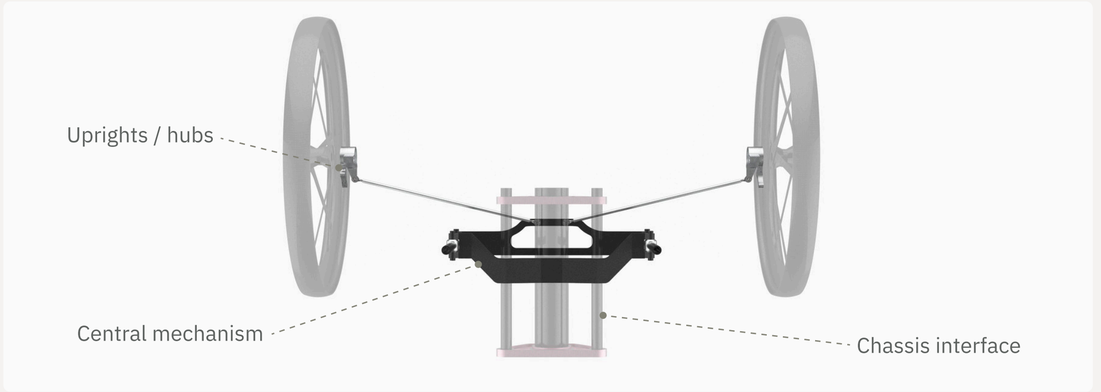
Led front-end steering development from mechanism architecture and CAD through prototyping, fabrication, testing, and integration.

- Mechanism architecture + CAD development
- Prototype iteration + tolerance / fit validation
- Mold design + carbon-fiber fabrication
- Final build, testing + subsystem integration

Four-person team. I led the steering-system technical development. **Adhvik Aggarwal** led project coordination, final rover assembly, and manufacturing oversight, while also contributing design feedback and metal fabrication. **Fangzhuo Tang** and **Raul Falcon** developed an alternate indirect-bearing steering concept. Carbon-fiber fabrication and final assembly were shared team responsibilities.

Steering architecture

CRANKER FOUR-BAR / FRONT-END SUBSYSTEM



Physical steering motion

Central mechanism → steering links → uprights / hubs → wheels

Four-Bar Steering Logic

- Top, bottom, and two side links form the steering linkage
- Crank input drives left/right steering through the connected bars

Design requirements / constraints

Geometry & clearance

Angle, bump steer, linearity / magnitude; ground, leg and A-arm clearance.

Precision & durability

Mechanical precision and durability; assembly and component stress testing.

Fabrication feasibility

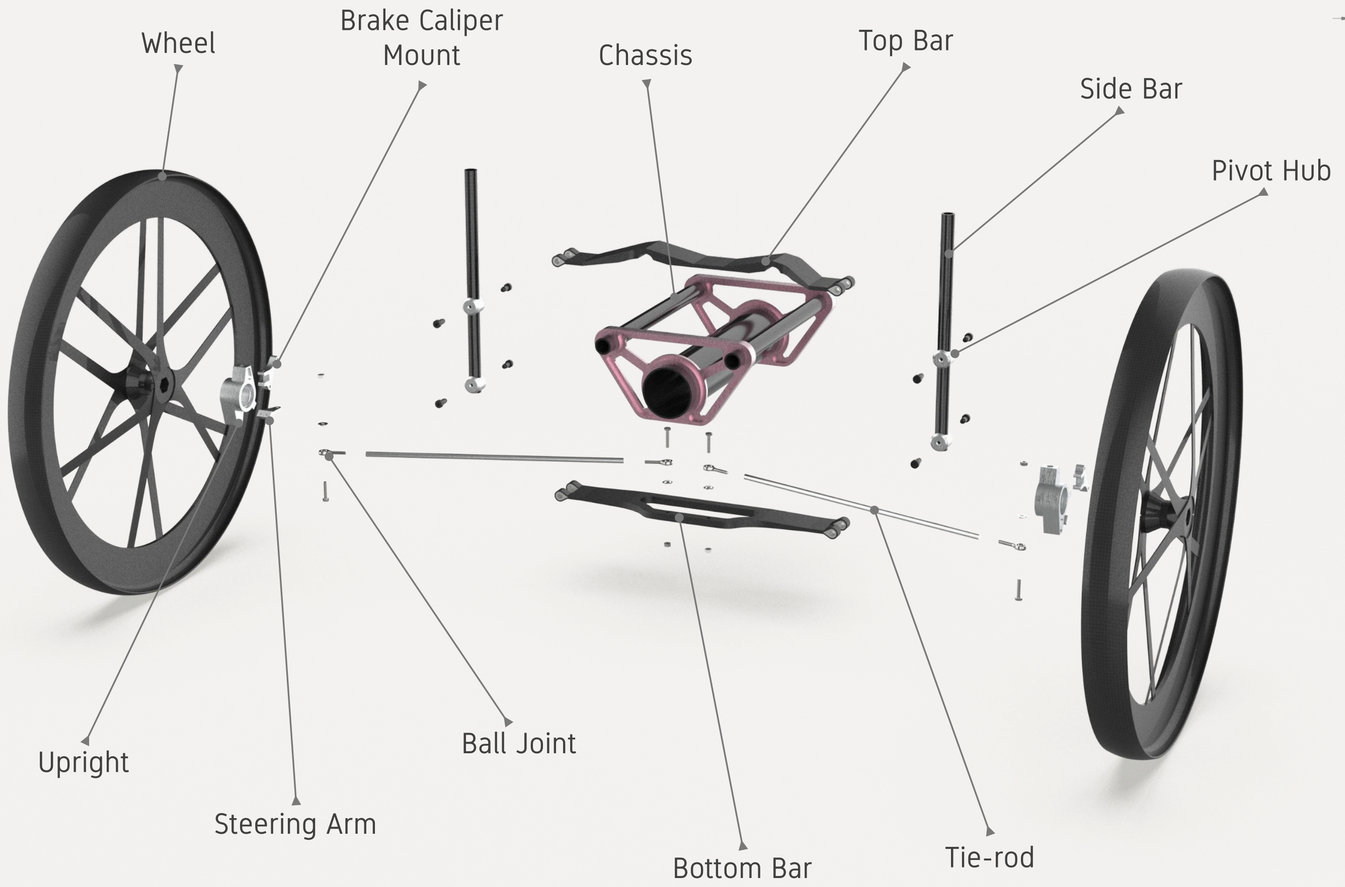
Molds + printed inserts; smooth transitions for carbon-fiber layup.

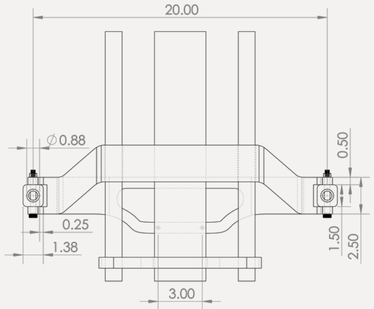
Subsystem integration

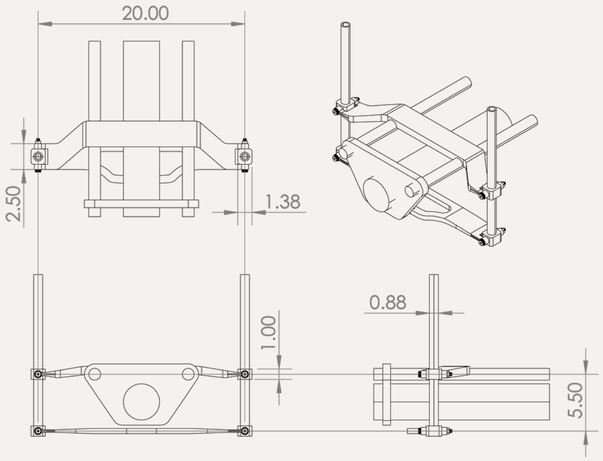
Chassis mounting + steering links; bolt-on validation and final integration.

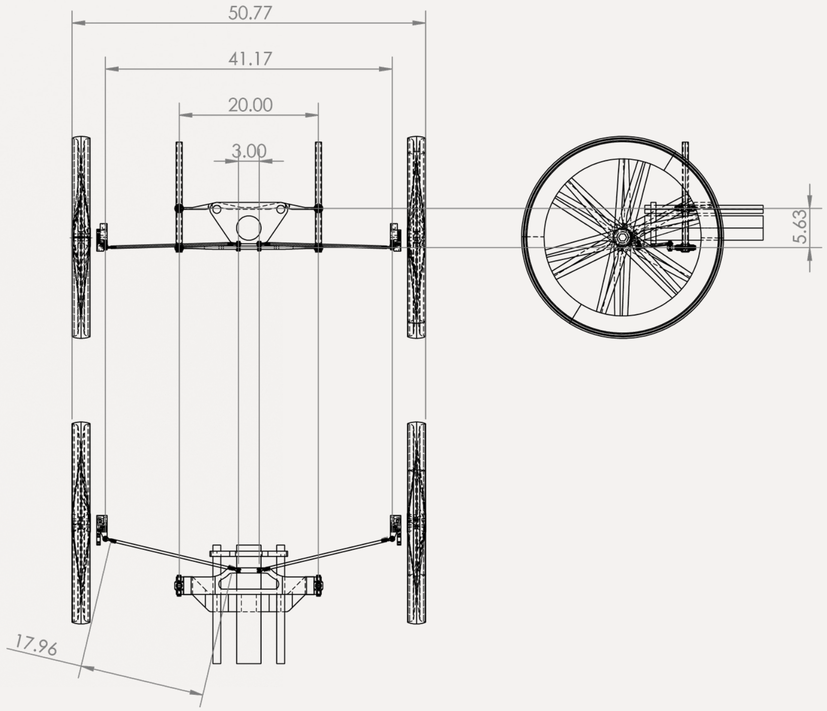
Mechanical Package

Exploded Assembly & Dimensions



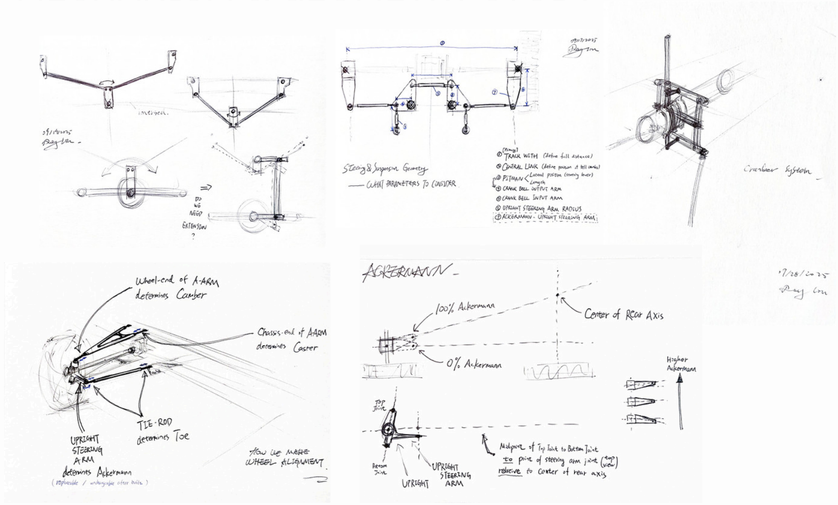






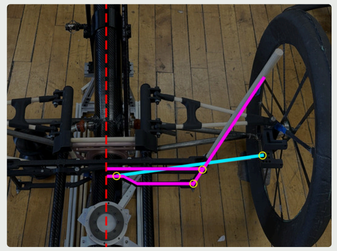
Development / Mechanism Decisions

MECHANISM & GEOMETRY STUDY

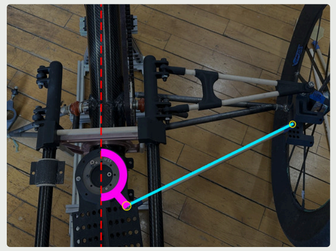


Early mechanism exploration:
linkage concepts and steering geometry.

DEVELOPMENT & DECISION



Cranker Four-Bar System



Indirect Bearing System

Early development compared two steering systems: **Cranker Four-Bar (C4B)** and **Indirect Bearing (IB)**.

The project moved forward with **Cranker Four-Bar** because the current packaging allowed better steering geometry, especially steering angle and bump steer.

Weight — IB = C4B

Bearing count vs. added supporting structure.

Sturdiness — IB > C4B

IB better resists unintended rider forces.

Intuitiveness — IB = C4B

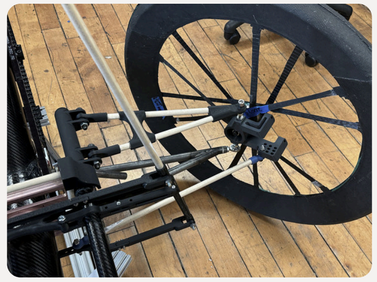
Bicycle familiarity vs. learnable crank input.

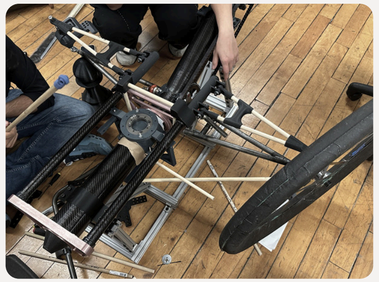
Steering Performance — IB < C4B

C4B enables cleaner linkage within this chassis.

SYSTEM & USER TESTING

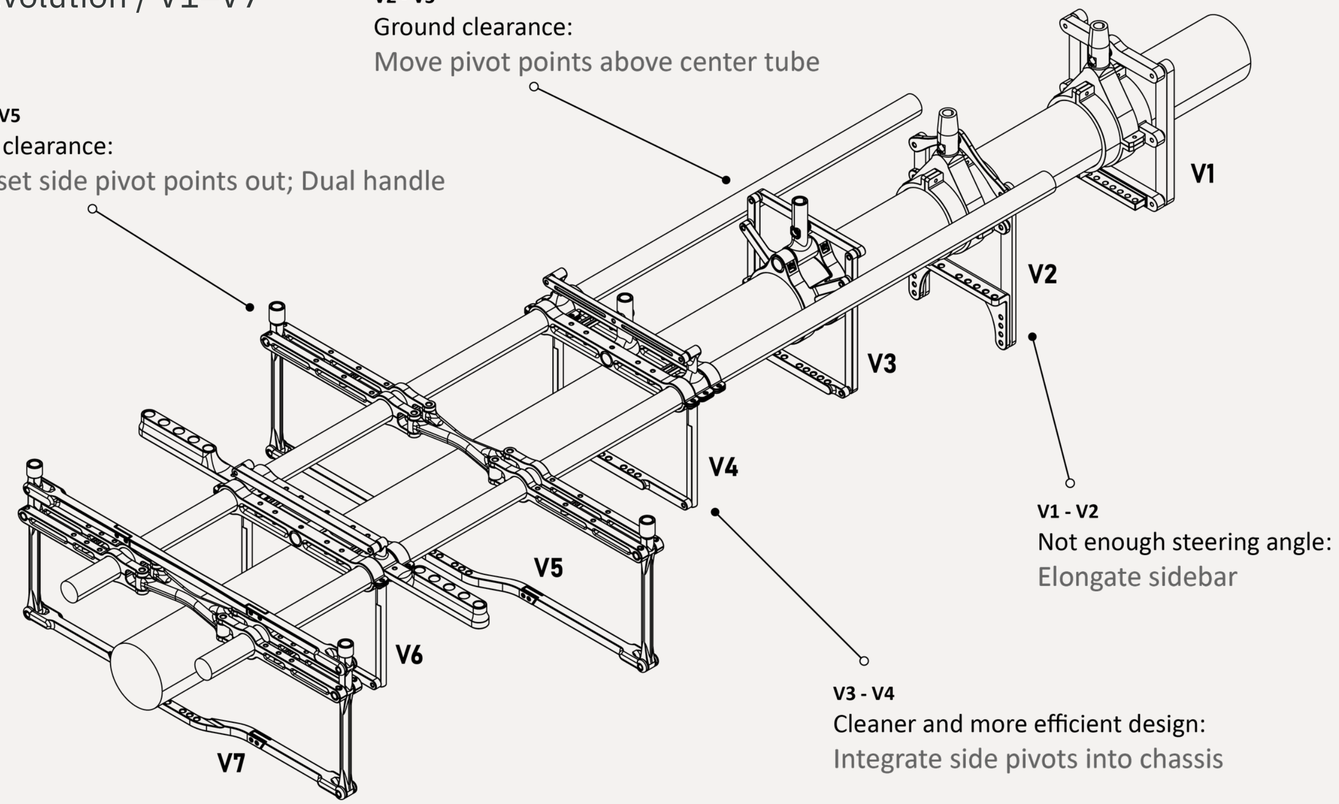
Physical trials compared how users interacted with the two systems: body position and movement, steering feel and feedback, controllability, and observed behavior.





Prototype / Iteration

System Evolution / V1–V7



V2 - V3
Ground clearance:
Move pivot points above center tube

V4 - V5
Leg clearance:
Offset side pivot points out; Dual handle

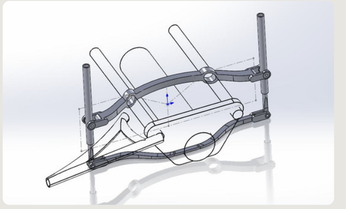
V1 - V2
Not enough steering angle:
Elongate sidebar

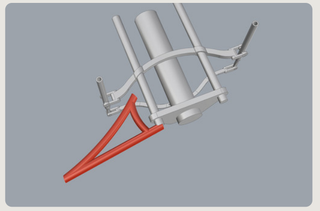
V3 - V4
Cleaner and more efficient design:
Integrate side pivots into chassis

3D printed bolt-on test supported repeated geometry and fit checks before final fabrication.

- Geometry + chassis integration
- Fit, tolerance + assembly
- Steering angle + bump steer
- Linearity + steering magnitude
- Ergonomics + clearance

A-arm conflict / Fab V1 → V2





Curve the center bar back; extend the bottom bar forward.

Fabrication Model Iteration

V2.0

Half clamp reduces complexity.

V3.5

Thicker transition near chassis tube for strength.

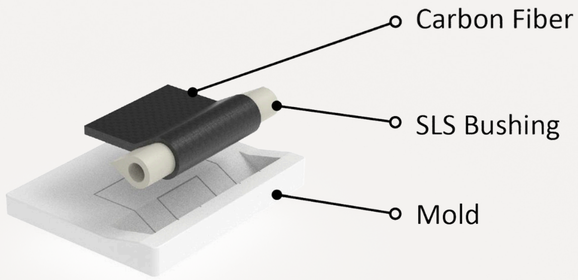
V3.75.5

Smooth carbon-to-print transition to reduce rupture risk.

V3.8.2

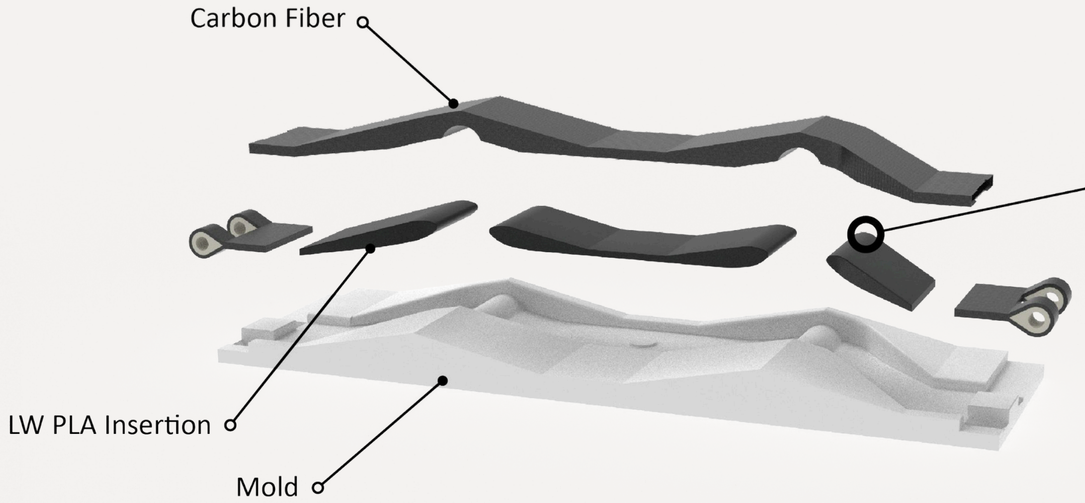
Fabrication / Mold Design

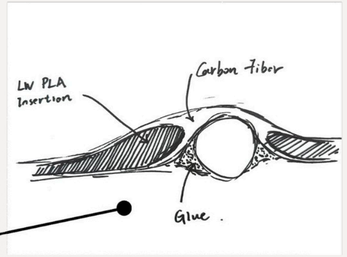
Mold Design & Iterations



End Flange

Version: End Flange Mold v6

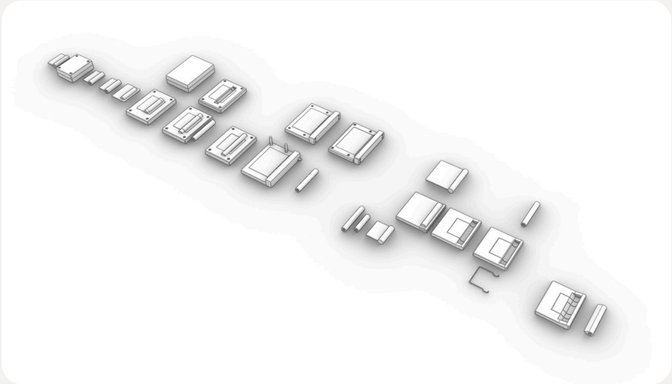




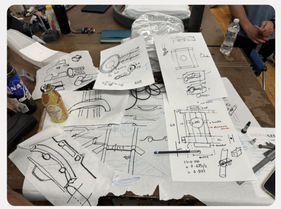
Top Bar

Version: Fab 3.8.2 TCP v3

End Flange Molds v1 - v6

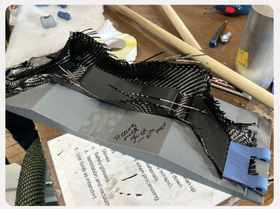


Fabrication & Stress Test



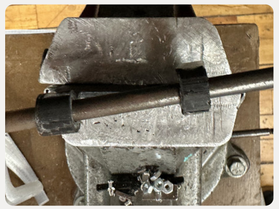
Fabrication concepts / dimensions

Bottom bar / mold test 1



Top bar v3 / carbon-fiber pre-layup

Top bar CF sample / stress test

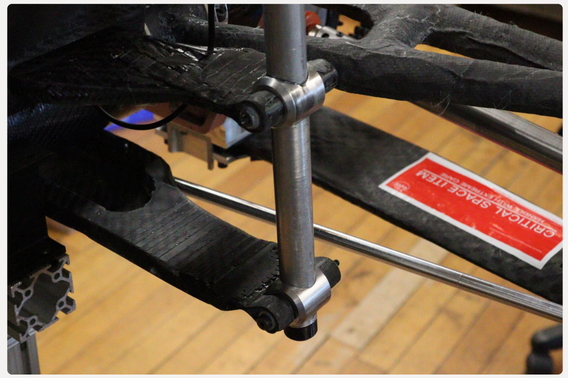


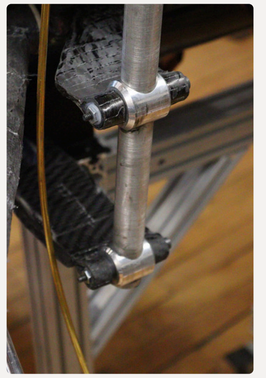
End flange / stress test + break

Bottom bar tie-rod attachment / stress test

Final System / Testing

Final Steering System

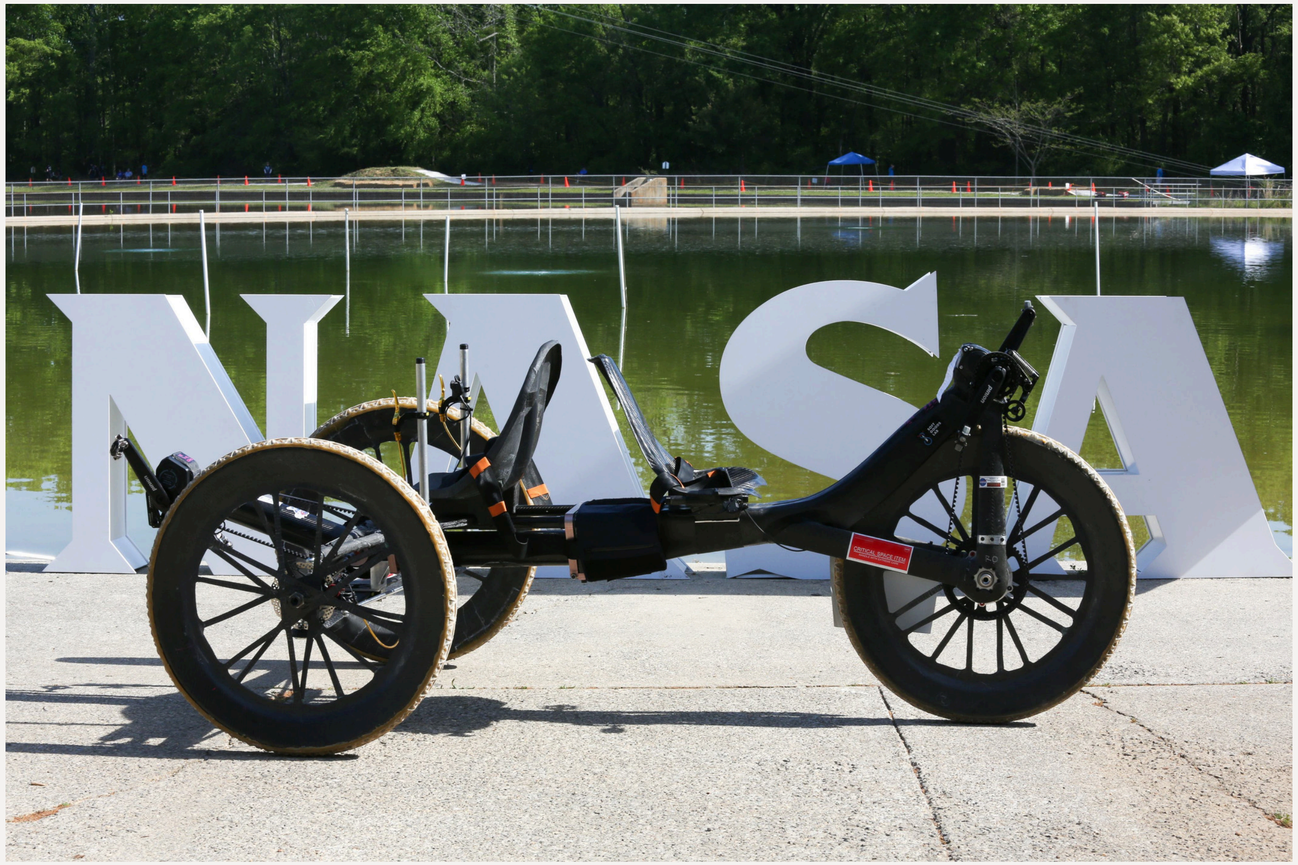




Outdoor ride testing evaluated steering response, controllability, mechanical robustness, and final subsystem integration.

Outdoor Testing

Competition / Outcome



2nd Place

2026 NASA Human Exploration Rover Challenge

Human Powered · College/University Division

Final Competition

Competition validation of the completed steering system under uneven-terrain conditions.

Thank you.

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